

## Chapter – 1: Geographical Diversity of India

To us, by its very geography, the country [India] appears to be quite distinct from other countries, and that itself gives it a certain national character. – Sri Aurobindo

- 1984 – Rakesh Sharma – 1<sup>st</sup> Indian astronaut – went to space – spoke with then Prime Minister – India Gandhi
- She asked – how does India look? – he replied – *Sare Jahan se Achchha*
- India – 7<sup>th</sup> largest – part of Asia
- Along with neighbours – Pakistan, Bangladesh, Nepal, Bhutan, Sri Lanka, Myanmar – forms region – Indian Subcontinent
- Purpose of discussion – refer to 5 regions – mountains, plains, desert, peninsula, islands
- This chapter – provide you – bird-eye's view – these features – offer glimpse – these places
- Natural borders –
  - Himalayan Mountain range – natural barrier – north
  - Thar desert, Arabian Sea – western limits
  - Indian Ocean – southern limits
  - Bay of Bengal – eastern limits
- These features – separation – India and continent – crucial role – shaping – India's climate, culture, history

### The Himalayas

- Look – length of Himalayan range – map – like a massive wall
- Summer – snow melts – feeds major rivers – Ganga, Indus, Brahmaputra
- These rivers, tributaries – provide water – drinking, farming, industrial use
- Himalayas – called – Water Tower of Asia – also imp. – many cultures, belief sys. – considered sacred
- Temples, monasteries – built in them – attracts monks, spiritual seekers – pray, meditate

### How the Himalayas were formed?

- Long time ago – India part of Gondwana – much larger landmass – neighbour – Africa
- Some point – broke away – started moving north
- 50 million yrs. ago – reached Eurasia – land in between – crumpled and rose up
- That's how Himalayan mountains formed
- India – still pushing into Asia – very slowly – 5 cm every year – slower than hair growth
- Himalayas – still growing – 5 mm each year – over a millennium – add upto 5 m
- Himalayas – broadly categorized – 3 main ranges –
  - **Himadri** –
    - Greater Himalayas – highest, most rugged
    - Towering peaks – Mt. Everest, Kanchenjunga
    - This region – snow-covered throughout
    - Life – very tough – not many human settlements
  - **Himachal** –
    - Lower Himalayas – south of greater Himalayas
    - More moderate climate – rich biodiversity – human habitation

- Popular hill stations – Nainital (Uttarakhand), Darjeeling (West Bengal), Shimla (Himachal Pradesh), Mussoorie (Uttarakhand)
- **Shivalik** –
  - Outer Himalayas – outermost, lowest range
  - Rolling hills – dense forest
  - Foothills – rich wildlife – transition zone b/w Himalayas and Gangetic Plains

## The cold desert of India

- Desert – gives an image – hot place
- India – cold deserts available – Ladakh – temp. – -30° C
- Very little rainfall – rugged landscape – rocky terrain – deep valleys – lakes – Pangong Tso
- Terrain here – resembles moon – moon land
- Geologists explain – formation – fact – mountains – folded – Indian landmass – collided with Eurasia
- Folded portion – part of ocean – rocks – this area – sand, clay
- Wind, rain – eroded mountains – in shape – as seen in pictures
- Harsh conditions – BUT – Ladakh – unique wildlife – snow leopards, ibex, Tibetan antelopes
- Ladakhi people – simple life
- This region – known for – ancient monasteries, colourful festivals – Losar, Hemis festival



## The Gangetic Plains

- South of Himalayas – vast, fertile Gangetic plains
- These plains – imp. part – history, civilisation – India
- These plains – nourished by – mighty rivers – originate from Himalayas – provide water
- Ganga, Indus, Brahmaputra – river sys. – along with – extensive network – tributaries – enrich soil – minerals – region – highly fertile, ideal – agriculture
- Rivers – also a src. – generate elec.
- Large portion – India's population – live here
- Flat land – Northern plains – allowed – development – elaborate transport network
- Road, rail network – helps movement – people, goods – long dist.
- Will see – Tapestry of the Past chapter – Ganga, Brahmaputra, other rivers – used – travel, trade

## The Great Indian Desert or Thar Desert

- Move west – notice – yellowish area – Thar desert
- Sand dunes – formed – wind shifts – shapes the sand – hill-like formations
- Sometimes – as high as 150 m
- The Thar – vast arid region – most of it – within India – spanning – Rajasthan, Gujarat, Punjab, Haryana
- Desert – natural barrier – reason – harsh conditions – deter human, animal movement – very high daytime temp. – cold nights – no access – water
- People in Thar – adapted – way of life – food habits, clothing, lifestyle – suited – harsh conditions
- Water – very scarce – women – travel long dist. – fetch water

## The Aravalli Hills

- Move a little East – Aravalli hills
- Aravallis – oldest mountains – 2.5 billion yrs. old
- This range – many peaks, ridges – highest peak – Mt. Abu – 1700 m
- BUT – most other hills – 300 - 900 m
- Very fascinating – 4.5 hrs. drive – Mt. Abu (Aravallis) to Jodhpur (Thar desert) – completely diff. geography
- Aravallis – vital role – shaping – geography, climate – northwestern India
- Most imp. func. – natural barrier – prevent – Thar desert – expanding further
- Aravallis – rich in minerals – marble, zinc, copper – supported mining, construction – centuries
- Evidence – ancient mines – Zawar – over 8 centuries ago – Indians – 1<sup>st</sup> – master the delicate process – extraction of Zinc
- Historic forts – Chittorgarh, Kumbhalgarh, Ranthambore – located here

## The Peninsular Plateau

- India – several plateau – most imp. – triangular peninsular area – middle, south of India
- This area – very old land formation
- This region – peninsula – surrounded by water – 3 sides – Arabian Sea, Bay of Bengal, Indian Ocean – known as peninsular plateau
- Plateau – guarded by 2 mountain ranges – Western, Eastern Ghats
- Western ghats – taller – run along western coast – many waterfalls – monsoon season
- Eastern ghats – lower – broken – smaller hills – along eastern coast
- Between these ranges – Deccan plateau – vast area – high flatlands
- Rivers – Godavari, Krishna, Kaveri – flow across plateau – west to east
- These rivers – imp. – farming, provide water
- Plateau – rich – minerals, forest, fertile land – vital – India's economy
- This plateau – tilts towards east – few rivers – flow into Bay of Bengal
- East-flowing rivers – Godavari, Krishna, Mahanadi – originate here – provide water – farming, industries, hydroelectric power
- Some rivers – west-flowing – Narmada, Tapi – drain into – Arabian Sea
- Dense forests – home – tribal communities – Santhal, Gond, Baiga, Bhil, Korku
- These tribes – distinct lang., traditions, ways of life – closely connected to nature
- Plateaus – home – many beautiful waterfalls – rivers flow over – uneven, rocky surfaces
- These waterfalls – attract tourists – provide water – irrigation, hydroelectric power plants

## India's Amazing Coastlines

- India's coastline – dotted – beautiful beaches, rocky cliffs, lush green forests
- Some beaches – golden sand – others – black rocks
- Some islands – coral reefs – others – covered in thick jungles
- Indian coastline – 7500+ km long

## West coast of India

- West coast – Gujarat to Kerala – passing through – Maharashtra, Goa, Karnataka

- Most rivers – originate here
- This coastline – alluvial deposits – short rivers – features – coves, creeks, estuaries
- Narmada, Tapi – estuaries largest
- This coast – many imp. ports, cities – centres of economic act.

### **The east coast**

- East coast – b/w Eastern Ghats and Bay of Bengal – Ganga delta to Kanyakumari
- Wide plains – major river deltas – Mahanadi, Godavari, Krishna, Kaveri
- Imp. water bodies – Chilika Lake, Pulicat lake – found here
- Deltas – landforms – mouth of river – deposits sediments into another water body – ocean, lake, another river
- Over time – sediments settle – form triangular, fan-shaped area
- All rivers – fertile deltas – land – ideal for farming

### **Indian Islands**

- Refer to – grp. of islands – scattered – Indian Ocean, Arabian Sea, Bay of Bengal
- India – 2 major island grps. – Lakshadweep (Arabian Sea) / Andaman and Nicobar Islands (Bay of Bengal)
- These islands – unique wildlife, beaches, coral reefs, volcanoes
- Many ancient tribes – these islands – their home

### **Lakshadweep islands**

- Archipelago (grp. of islands) – Arabian Sea – close to – Malabar Coast of Kerala
- Made of 36 islands of coral – not all inhabited
- India – controls – vast marine area – fishing, resource exploration, environmental protection, etc.

### **Andaman and Nicobar Islands**

- This archipelago – 500+ volcanic islands – 2 distinct grps. – Andaman Islands, Nicobar Islands
- These islands – very imp. – they are like outposts – keeping eye on ocean – home – variety – flora, fauna
- Andaman Islands – significant – historic view – many freedom fighters – jailed there – most severe conditions – cellular jail
- This jail – preserved – remind us – tremendous sacrifice – make our country free

### **Delta in West Bengal and Sundarbans**

- Travel back from islands – toward eastern side – via Bay of Bengal – reach Sundarbans
- Located – delta – Ganga, Brahmaputra and their tributaries
- This delta – unique combo. – river, sea, land – half of this – India – rest half – Bangladesh
- This place – UNESCO heritage site – home to many species – including Royal Bengal Tiger

### **Hills of the Northeast**

- Move towards hills – northeast – Garo, Khasi, Jaintia – part of Meghalaya Plateau
- These hills – known for – lush greenery, heavy rainfall, waterfalls
- This region – highest rainfall – making it rich – forests, wildlife, fertile land